Swastik Mittal

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Summary

PhD candidate researching **timing integrity and predictability of real-time operating systems in high-performance computing**. Pursuing a career as a **systems and performance engineer** — with hands-on expertise in **compiler optimization, GPU scheduling/optimization, parallel computing, Linux-based networks and real-time safety**, backed by publications and production deployments.

Publications

- **OpenMP-RT-Offload**: Real-Time Support for OpenMP GPU Offloading — Under Submission
- **OpenMP-Q**: Quantum Task Offloading Extension for OpenMP — IWOMP 2025
- **T-Tex**: Timed Security in Multi-Threaded Real-Time Systems — ICCPS 2025
- **T-Pack**: Timed Network Security for Real-Time Systems — ISORC 2021

Education

PhD, Computer Science — North Carolina State University, Raleigh, NC
Compiler Optimization, Embedded/Real-Time Systems, Parallel Systems — *Aug 2020 – Present*

MS, Computer Science — North Carolina State University, Raleigh, NC
Operating Systems, Linux Networking, Internet Protocol, Graph Theory — *Aug 2018 – Apr 2020*

B.Tech., Computer Science — Vellore Institute of Technology, India
Parallel and Concurrent Systems, Cloud Computing, Agent Based Intelligent Systems — *Jul 2014 – Apr 2018*

Work Experience

AIM Intelligent Machines — Monroe, WA
Performance Optimization Engineer — *May 2024 – Aug 2024*

- Reduced CPU process latency by $\geq$80% in a real-time autonomous robotics pipeline (Autonomous Dozer) through targeted profiling (Perf, Nsight Compute, isolated execution analysis).
- Implemented MPS-integrated GPU scheduling framework, cutting GPU latency step time by $\approx$9% in the pipeline.

Programming Systems, Uber — San Francisco, CA
Compiler Optimization Engineer — *May 2022 – Aug 2022*

- Implemented profile-guided code layout optimization (PGO) in the Go linker using a cross-package weighted call graph with C3 heuristics
- Optimized function layout for cache utilization across Google-scale multi-package Go benchmarks

Research Experience

Graduate Research Assistant — North Carolina State University, Raleigh, NC
Aug 2019 – Present

Research Advisor: Dr. Frank Mueller
OpenMP-Q: Quantum Task Offloading for HPC (IWOMP 2025):

- Designed an OpenMP language extension integrating quantum task offloading into HPC workflows via POSIX pipes, achieving up to **2.9× speedup**
- Built a quantum-circuit library supporting reverse offloading and broad hardware compatibility

OpenMP-RT-Offload: Real-Time GPU Scheduling for OpenMP (Under Submission):

- Designed an application-transparent interposer using NVIDIA Green Contexts and per-partition priority-inheritance queues to deliver real-time scheduling for OpenMP GPU offloading without modifying application code or the OpenMP runtime
- Reduced worst-case response time by **12–25ms** for high-priority tasks vs. baseline
- For 8 tasks over 2 partitions: **59%** miss rate reduction vs. baseline, **50%** vs. state-of-the-art, and **zero misses** for the second-highest-priority task

T-Tex: Timed Security in Multi-Threaded Real-Time Systems (ICCPS 2025):

- Developed OpenMP code region identification using LLVM/Clang to detect and prevent timing-based delay attacks in real-time systems
- Designed a kernel-to-runtime communication interface to monitor context switches and secure timers, enabling worst-case execution time (WCET) evaluation
- Achieved **100% attack detection** for 60μs delay attacks with minimal runtime overhead on an autonomous driving benchmark

T-Pack: Network Security for Safety Critical Real-Time Systems (ISORC 2021):

- Built a Linux kernel module using socket buffers for real-time transmission timing analysis, detecting **95–100% of attacks** with $\approx$0.012ms overhead
- Extended to all major network protocols with IPSec encryption compatibility

Advisor: Dr. Hung-Wei Tseng
In-Network Processing Analysis:

- Developed Apache Spark/Hadoop benchmarks to identify network congestion bottlenecks in SDN/5G edge computing environments

Teaching Experience

Graduate Teaching Assistant — Parallel Systems, NC State University

- Mentored graduate students on CUDA/NVIDIA GPU programming, MPI cluster programming, OpenMP/OpenACC multithreading, and system performance/power optimization

Projects

Autonomous Vehicle Simulator (Carla):

- Designed a PID controller and CNN model to simulate autonomous driving; **100% of 10 concurrent vehicles** successfully followed traffic signs

Cloud Provider with DNS-as-a-Service:

- Built a virtual private cloud using Linux LXD containers with hierarchical DNS (bind9) and load balancing as default services

Peer-to-Peer File Sharing System:

- Implemented a dynamic TCP/UDP file sharing system with stop-and-wait reliability for point-to-multipoint transfer, achieving **95% network efficiency**

Skills

Languages: C, C++, Python, Go

High Performance Computing: OpenMP, MPI, CUDA, OpenACC

Compilers & Optimization: LLVM/Clang, DynamoRIO, Go Compiler, PGO, Profile-Guided Optimization

Systems Programming: Linux Kernel Modules, Network Stack, Process/GPU Scheduling, Signals, Timers, Real-Time Inference, Safety-Critical Systems, CNN, PID Controller, Carla Simulator, Autonomous Driving Pipeline

Distributed & Networked Systems: DNS Service, Virtual Private Cloud, TCP/UDP Protocols, Linux Network Stack, Apache Spark, Hadoop, SDN, IPSec, KVM, CloudStack